MemMapGen Error Description V12.7.0

Error Description Help Document

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ETAS MemMapGen 12.7.0 - MemMapGen Error Description Help Document R01 EN-05.2025

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1 Safety and Privacy Information

1.1 Intended Use

This document can be referred to get the information and resolution of the error, warning and info messages logged from MemMapGen.

1.2 Target Group

All ETAS RTA-CAR users who wants to generate memory mapping header files for AUTOSAR components.

1.3 Classification of Safety Messages

The safety messages used here warn of dangers that can lead to personal injury or damage to property:

DANGER

DANGER indicates a hazardous situation that, if not avoided, will result in death or serious injury.

WARNING

WARNING indicates a hazardous situation that, if not avoided, could result in death or serious injury.

CAUTION

CAUTION indicates a hazardous situation that, if not avoided, could result in minor or moderate injury.

NOTMEMMAPGEN

NOTMEMMAPGEN indicates a situation that, if not avoided, could result in damage to property.

1.4 Safety Information

This document is only for the information and suggested resolutions of the error, warning and info messages logged from MemMapGen.

1.5 Privacy

Made available with RTA-CAR lMemMapGennse

1.6 Definitions and Abbreviations

Term/Abbreviation	Definition
HW	HardWare
SW	SoftWare
ASW	AUTOSAR Application SoftWare

Term/Abbreviation	Definition
BSW	AUTOSAR Basic SoftWare

2 About this Manual

This document provides a list of validations error messages. For each error message a description of the cause and possible solutions are included.

3 MemMapGen Validations

This section deals with the error description of MemMapGen validations. This is applicable for all the MemMapGen AUTOSAR elements.

3.1 AUTOSAR memory mapping generator validations

This section describes about MemMapGen general validations.

3.1.1 AR_MEMMAP_ERROR_001

3.1.1.1 Error Description

Missing or empty <SHORT-NAME> in the <MEMORY-SECTION> of the input RTE BSW module arxml file containing memory sections.

3.1.1.2 How to Solve

Check <SHORT-NAME> tag present inside <MEMORY-SECTION>. If present, check <SHORT-NAME> text which should not be empty. If it is empty or missing, then analyze why RTE tool is generating arxml file with empty/missing shortName. or any manual changes done on arxml file.

3.1.1.3 Error Message Syntax

Missing or empty <SHORT-NAME> in the <MEMORY-SECTION> of the arxml file {{filePath_FILE_PATH}}. Corresponding element is not considered.

3.1.2 AR_MEMMAP_INFO_001

3.1.2.1 Error Description

The Vendor Id and Vendor API are expected to be available in the same *. arxml file. If Id and API are not found in the same *. arxml file, then this info will be logged.

3.1.2.2 How to Solve

Component Name generation for different <vi> and <ai> for same component.

3.1.2.3 Error Message Syntax

The Vendor Id and Vendor API are available in different input files, <VENDOR-ID> and <VENDOR-API-INFIX> will not be added to the component name "{{componentName}}".

3.1.3 AR_MEMMAP_ERROR_010

3.1.3.1 Error Description

Memory sections can be defined in MemMapGen templates as shown below

«addMemorySection("TEST")»

«addMemorySection("TEST")»

or

«addMemorySection("TEST_RTE","RTE")»

«addMemorySection("TEST_RTE","RTE")»

If duplicate sections are added, then the tool throws this error and considers only first definition defined.

3.1.3.2 How to Solve

Memory section which is not required can be deleted.

3.1.3.3 Error Message Syntax

Duplicate <MEMORY-SECTION> "{{MemorySectionName}}" found -> used in the extend utility API {{extendAPIname}}. Only first <MEMORY-SECTION> is considered.

3.1.4 AR_MEMMAP_ERROR_011

3.1.4.1 Error Description

When it is tried to add memory section from MemMapGen templates in the RTE component and that component is not present in the input arxml files. Then this error is thrown.

«addMemorySection("TEST_RTE","RTE”)»

3.1.4.2 How to Solve

Configure the memory section in MemMapGen template with existing RTE component to which memory section should be created.

3.1.4.3 Error Message Syntax

Unable to add <MEMORY-SECTION> "{{MemorySectionName}}", as Component: "{{ComponentName}}" is not found in input arxml files.

3.1.5 AR_MEMMAP_WARN_001

3.1.5.1 Error Description

If an address method is referred in memory section and that address method is not defined in the AUTOSAR arxml file, then this warning is logged.

3.1.5.2 How to Solve

Can be verified if the address method definition referred in memory section is required or not.

3.1.5.3 Error Message Syntax

The <{{Element}}> element {{ElementName}} in the file {{filePath_FILE_PATH}} is having <{{ElementRef}}> which is not defined. Hence this element is not considered.

3.1.6 AR_MEMMAP_WARN_009

3.1.6.1 Error Description

Memory sections can be defined in MemMapGen templates as shown below

«addMemorySection("TEST")»

«addMemorySection("TEST_RTE","RTE")»

If this sections are configured with empty string example: «addMemorySection("")», then this error is logged.

3.1.6.2 How to Solve

If required to define memory sections in XPT(MemMapGen template) files, memory sections needs to be configured correctly as per the template.

3.1.6.3 Error Message Syntax

Missing or empty <SHORT-NAME> for the <MEMORY-SECTION> used in the extend utility API -> {{extendAPIname}}.

3.1.7 MEMMAPGEN_XPD_PLATFORM_0002

3.1.7.1 Error Description

Platform templates does not define the version of the templates.

3.1.7.2 How to Solve

Define the version information for the templates.

Please refer to MemMapGen Reference Manual for more details.

3.1.7.3 Error Message Syntax

No {{templateType}} template defines the version info for {{templateType}} {{versionName}}.

3.1.8 AR_MEMMAP_INFO_002

3.1.8.1 Error Description

If the <Module>_BSWMD.arxml file and Customer Templates does not contain <MEMORY-SECTIONS> then the <Module>_MemMap.h is not generated.

3.1.8.2 How to Solve

This can be verified if the <MEMORY-SECTIONS> tag can be included in the <Module>_BSWMD.arxml to generate the particular <Module>_MemMap.h file and if addMemorySection() is present in Customer Templates.

3.1.8.3 Error Message Syntax

The file "{{filePath_FILE_PATH}}" doesn't contain <MEMORY-SECTIONS> and addMemorySection() is not configured in CustomeRTEmplates, so the corresponding component memmap.h file is not generated.

3.1.9 AR_MEMMAP_WARN_003

3.1.9.1 Error Description

Duplicate memory sections with same shortName are defined in the RTE BSW module arxml files.

3.1.9.2 How to Solve

Analyze why RTE tool is generating memory sections in arxml files with same name. Or any manual changes done on arxml file.

3.1.9.3 Error Message Syntax

Duplicate <MEMORY-SECTION> "{{MemorySectionName}}" in the arxml files {{filePath_FILE_PATH}}. Only first <MEMORY-SECTION> is considered from {{filePath_FILE_PATH}}.

3.1.10 AR_MEMMAP_ERROR_002

3.1.10.1 Error Description

If input arxml doesn't consist of short name in BswModuleDescription/SwComponentType, then tool will abort.

3.1.10.2 How to Solve

Component shortName Should be added to the respective component BSWMD/SWCD arxml files.

3.1.10.3 Error Message Syntax

Missing short name in the <BSW-MODULE-DESCRIPTION>/<SW-COMPONENT-TYPE> of input file {{filePath_FILE_PATH}}.

3.1.11 AR_MEMMAP_WARN_007

3.1.11.1 Error Description

If component has same vendor APIs but different vendor Id, then tool this warning will be logged.

3.1.11.2 How to Solve

Component Name generation for different <vi> and <ai> for same component.

3.1.11.3 Error Message Syntax

The component "{{componentName}}" has different Vendor Ids, Vendor Id and Vendor API will not be appended to component name.

3.1.12 AR_MEMMAP_WARN_008

3.1.12.1 Error Description

Warnings that are generated during processing of RTATEXT templates.

3.1.12.2 How to Solve

To be reviewed and fixed by checking the message content.

3.1.12.3 Error Message Syntax

Schema Validation warning: {{message}} at line number {{line}}, column number {{column}} and file "{{filePath}}".

3.1.13 AR_MEMMAP_ERROR_006

3.1.13.1 Error Description

When the component short name is in different case in multiple *arxml files for same component name.

3.1.13.2 How to Solve

Enhancement AR MemMapGen for camel case component names.

3.1.13.3 Error Message Syntax

The component short name "{{componentName}}" is in different case in one of the input ARXML files {{filePath_FILE_PATH}}.

3.1.14 AR_MEMMAP_WARN_010

3.1.14.1 Error Description

If prefixRef is present and symbol is missing in the memorySection, then the warning message would be logged. For generating memory allocation keywords based on prefix ref and symbol, the memory section should consist of both.

3.1.14.2 How to Solve

Generating memory allocation keywords based on the PREFIX-REF and SYMBOL tags.

3.1.14.3 Error Message Syntax

<SYMBOL> tag is missing in <MEMORY-SECTION> '{{memorySectionName}}'. The memory allocation keywords would be generated based on BswModuleDescription.shortName and MemorySection.shortName.

3.1.15 AR_MEMMAP_WARN_011

3.1.15.1 Error Description

According to RTE specifications, the prefixRef should start with componentName.

3.1.15.2 How to Solve

Generating memory allocation keywords based on the PREFIX-REF and SYMBOL tags.

3.1.15.3 Error Message Syntax

According to RTE specifications, <PREFIX-REF> '{{prefixRef}}' in memorySection name '{{memorySectionShortName}}' should start with component name '{{compName}}'.

3.1.16 AR_MEMMAP_WARN_012

3.1.16.1 Error Description

If the number of groups in the memory section name having regex in it and the number of replacement strings in the pragma string does not match, then this warning message will be logged.

3.1.16.2 How to Solve

The groups in memory section name and the replacement patterns in the pragma string should match.

3.1.16.3 Error Message Syntax

Number of groups in the pattern "{{pattern}}" does not match the number of $x(replacement string) in replacement pattern.

3.1.17 AR_MEMMAP_WARN_013

3.1.17.1 Error Description

This message will be logged when a reference tag is missing or when the reference tag is available but the definition for the referenced element is missing.

3.1.17.2 How to Solve

Generating the MemMapGen file only when the reference tag and the definition for the referenced element available.

3.1.17.3 Error Message Syntax

The <{{Element}}> element {{ElementName}} in the file {{filePath_FILE_PATH}} {{having/not having}} <{{ElementRef}}> {{which is not defined}}. Hence the {{filePath_FILE_PATH}} will not be considered for MemMapGen generation.

3.1.18 AR_MEMMAP_WARN_014

3.1.18.1 Error Description

Xpand and RTATEXT Templates are passed for MemMap generation.

3.1.18.2 How to Solve

Pass only RTATEXT Templates for MemMap generation.

3.1.18.3 Error Message Syntax

Xpand and RTATEXT Pragma templates are passed. Xpand Templates are ignored, only RTATEXT Templates are considered for MemMap generation.

3.1.19 AR_MEMMAP_WARN_006

3.1.19.1 Error Description

This message will be logged for the missing pragma section configurations for the memory sections present in the input arxml file.

3.1.19.2 How to Solve

Check memory section names in input arxml. When the pragma section isn’t generated for the memory-section, then the compiler default pragma section is considered during linking.

3.1.19.3 Error Message Syntax

Pragma configuration couldn't be found for the <MEMORY-SECTION> '{{memorySection-Name}}' present in the arxml file {{filePath_FILE_PATH}}.

3.1.20 ARA2L_ARTOP_WARN_001

3.1.20.1 Error Description

If the input arxml is having warnings according to AUTOSAR schema file, then this warning message is logged with details. But the tool still processes the files and ignores the warnings in arxml file.

3.1.20.2 How to Solve

Check if the warnings really cause any problem. otherwise, these can be ignored.

3.1.20.3 Error Message Syntax

Schema Validation warning: {{message}} at line number {{line}}, column number {{column}}.

3.1.21 ARA2L_ARTOP_ERROR_001

3.1.21.1 Error Description

If the input arxml is not well formed according to AUTOSAR schema file, then this error is logged with details and the tool aborts after logging the next error ARA2L_ARTOP_ERROR_001.

3.1.21.2 How to Solve

Analyze why RTE tool is generating arxml file which is not well formed according to the AUTOSAR schema. Or any manual changes done on arxml file.

3.1.21.3 Error Message Syntax

Well formness error: {{message}} at line number {{line}}, column number {{column}}.

3.1.22 ARA2L_ARTOP_WARN_002

3.1.22.1 Error Description

If the input arxml is not valid according to AUTOSAR schema file, then this error is logged with details. But the tool still processes the files and ignores the error present in arxml file.

3.1.22.2 How to Solve

Check the error details for line numbers to know where exactly the problem in input arxml file is. If the data is required, then analyze why RTE is generating data which is not according to the AUTOSAR schema.

3.1.22.3 Error Message Syntax

Schema Validation error: {{message}} at line number {{line}}, column number {{column}}.

3.1.23 ARA2L_ARTOP_ERROR_002

3.1.23.1 Error Description

If the input arxml is not well formed according to AUTOSAR schema file, then this error is logged with details and the tool aborts after logging the next error.

3.1.23.2 How to Solve

Analyze why RTE tool is generating arxml file which is not well formed according to the AUTOSAR schema. Or any manual changes done on arxml file.

3.1.23.3 Error Message Syntax

Aborting due to presence of well formness error.

3.1.24 ARA2L_ARTOP_ERROR_003

3.1.24.1 Error Description

If the input ARXML file does not exists.

3.1.24.2 How to Solve

Provide the correct input arxml file path.

3.1.24.3 Error Message Syntax

The McSupportFile '{{mcSupportFileName}}' does not exist.

3.1.25 AR_MEMMAP_ERROR_003

3.1.25.1 Error Description

This message will be logged for the missing symbol tag in Section-Name-Prefix for the memory sections present in the input arxml file.

3.1.25.2 How to Solve

Provide symbol tag for the Section-Name-Prefix.

3.1.25.3 Error Message Syntax

SYMBOL tag is missing in '{{prefixRef}}'. SYMBOL is a mandatory parameter for SECTION-NAME-PREFIX.

3.1.26 AR_MEMMAP_ERROR_004

3.1.26.1 Error Description

This message will be logged for the missing pragma configurations in the template files for available memory sections.

3.1.26.2 How to Solve

provide pragma configurations for the available memory sections in the templates.

3.1.26.3 Error Message Syntax

Pragma configurations are missing in the templates for the available memorySections in the input arxml file list '{{path}}'.

3.1.27 AR_MEMMAP_ERROR_005

3.1.27.1 Error Description

Critical Errors which are generated during processing of RTATEXT templates.

3.1.27.2 How to Solve

Tool aborts, So, to be reviewed and fixed immediately by checking the message content.

3.1.27.3 Error Message Syntax

Schema Validation error: {{message}} at line number {{line}}, column number {{column}} and file "{{filePath}}".

3.1.28 MEMMAPGEN_SYSTEM

3.1.28.1 Error Description

This is a system level message and does not correspond to any issue.

It is just an information message.

3.1.28.2 How to Solve

None

3.1.28.3 Error Message Syntax

{{SYSTEM_MESSAGE}}

3.1.29 MEMMAPGEN_DEBUG_RUNTIME

3.1.29.1 Error Description

This is an info message for the time taken by MEMMAPGEN tool.

3.1.29.2 How to Solve

None.

3.1.29.3 Error Message Syntax

Total time taken for {{Component name}}: {{Time ms}}ms ({{Time s}}s).

3.1.30 MEMMAPGEN_SYSTEM_RUNTIME

3.1.30.1 Error Description

This is an info message for the time taken by MEMMAPGEN tool.

3.1.30.2 How to Solve

None.

3.1.30.3 Error Message Syntax

Total time taken for {{Component name}}: {{Time ms}}ms ({{Time s}} s).

3.1.31 MEMMAPGEN_VERSION

3.1.31.1 Error Description

This is just an information message about the MEMMAPGEN version.

3.1.31.2 How to Solve

None.

3.1.31.3 Error Message Syntax

Generated by MEMMAPGEN Version: {{MEMMAPGEN_PROD_VERSION}}.

3.1.32 MEMMAPGEN_XPD_0001

3.1.32.1 Error Description

There are more than one xpand definition with same name defined in either platform or customer templates.

3.1.32.2 How to Solve

Remove all the duplicate definitions and only retain one of them.

3.1.32.3 Error Message Syntax

Multiple definitions of Xpand definition: '{{Xpand Definition Name}}' is found in {{templat-etype}} template files: '{{file1_FILE_PATH}}' , '{{file2_FILE_PATH}}'.

3.1.33 MEMMAPGEN_XPD_0006

3.1.33.1 Error Description

The impoRTEd extension 'xtendFileName' in the xpand template 'xptFileName' does not exist in 'templateFolderPath', where the xpand template exists.

3.1.33.2 How to Solve

- The impoRTEd extension (*.ext) and the xpand template (*.xpt) should be present in the same folder.
- Copy the impoRTEd extension to 'templateFolderPath'.

3.1.33.3 Error Message Syntax

{{xptFileName}}: The impoRTEd extension {{xtendFileName}} does not exist in template resource folder: {{templateFolderPath_FILE_PATH}}.

3.1.34 MEMMAPGEN_XPD_0007

3.1.34.1 Error Description

The specified template does not exist in the disk.

3.1.34.2 How to Solve

Copy the specified template to the template folder location.

3.1.34.3 Error Message Syntax

The specified template path {{templatePath_FILE_PATH}} does not exist.

4 Contact Information

Technical Support

For details of your local sales offMemMapGen as well as your local technical support team and product hotlines, take a look at the website: www.etas.com/hotlines

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